

PAPER_1331 — Pop III Top-Heavy IMF Characteristic Mass

REFINED June 2026: $M_{\text{PopIII}} = A_5 \times (D_{\text{phys}}+1)/(D_{\text{phys}}-1) = 60 \times 5/3 = 100 M_{\text{sun}}$
EXACT

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "pop_iii_imf"})`

Master Expression

$$M_{\text{PopIII}} = A_5 \times 2 = 120 M_{\text{sun}}$$

Match: 20% from observed ~100

Interpretation

First stars top-heavy via unenriched SCm buoyancy at $z > 15$. JWST consistent.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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